We have

$$\begin{aligned}
f(x) &= \frac{x^2}{100} - 2x + 1, \\
f'(x) &= \frac{2x}{100} - 2, \\
f''(x) &= \frac{2}{100}.
\end{aligned}$$

We need to find M such that

$$|f'(x) - f'(y)| \leq M|x - y|.$$

We have

$$|f'(x) - f'(y)| = \left| \frac{2x}{100} - \frac{2y}{100} \right| = \frac{2}{100}|x - y|.$$

Therefore, any M such that

$$M \geq \frac{2}{100}$$

is a Lipschitz constant.